COMPARISON OF THE PH AND ETALE TOPOLOGIES: A_X^{-1} AND THE VANISHING OF R^I EPSILON_{X,*}

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-comparison-ph-etale (source lines 1888-1898); the proof environment beginning on the line after the following blank line (source line 1900) is its own proof and closes at source line 1955.

Editorial note. The proof invokes two results of the same chapter by label, both retained as context with the generator's transitive expansion; it also cites four cross-chapter results (Etale Cohomology twice, Cohomology on Sites twice), each declared as a tagged source pointer.

Editorial note. The author writes that part (2) is proved by reduction to schemes and that 'The details are given below but we urge the reader to skip the proof.'; the remark is reproduced verbatim and nothing is added in its place.

Editorial note. The retained preamble is the chapter's own (sources/stacks-spaces-more-cohomology/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup and theorem declarations; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's chapter Cohomology of Algebraic Spaces, counted once: the comparison of the ph and etale topologies, namely the identities for a_X^{-1} and the vanishing of the higher direct images of epsilon_{X,*} on torsion sheaves.

Allowed standard prerequisites (editorial).

- The maps between the small and big toposes. a_X and epsilon_X with a_X^{-1}F = epsilon_X^{-1}pi_X^{-1}F, and the fact that epsilon_{X,*} is the identity on underlying presheaves while pullback by epsilon_X is ph sheafification.. It is the starting point of part (1) at source lines 1901-1907.
- The description of pi_X^{-1}F and its sheaf property. pi_X^{-1}F is computed by Gamma(Y_etale, f_small^{-1}F) and is a sheaf for the ph (and smooth, syntomic, fppf, fpqc) topology (the author's lemma-describe-pullback, retained as context).. It gives epsilon_{X,*}a_X^{-1}F = pi_X^{-1}F at source lines 1902-1909.
- Higher direct images as associated sheaves of cohomology presheaves. For an abelian sheaf H on (Spaces/X)_ph the sheaf R^p epsilon_{X,*}H is the sheafification of U -> H^p_ph(U,H) on the small etale site (Cohomology on Sites, declared as a tagged source pointer).. It turns the vanishing into a statement about classes over schemes at source lines 1920-1929.

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- Cohomology agrees over (Spaces/U)_ph and (Sch/U)_ph. The restriction identification H^p((Spaces/U)_ph, -) = H^p((Sch/U)_ph, -) used for a scheme U (the author's lemma-compare-cohomology-other-topologies, retained as context).. It is the second identification of the displayed chain at source lines 1937-1938.
- The scheme case of the comparison. The corresponding statements for schemes: the description of the pullback in the schemes version and V_n for all n, i.e. the vanishing in the scheme case (Etale Cohomology, declared as tagged source pointers).. It finishes part (2) at source lines 1941-1952.

Source reference. Original lemma 0DE4: sources/stacks-chow/tags.txt, source lines 16789–16789. Cross-chapter pointer whose target carries the canonical tag 0DE4 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 072W: sources/stacks-chow/tags.txt, source lines 9040–9040. Cross-chapter pointer whose target carries the canonical tag 072W in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03F3: sources/stacks-chow/tags.txt, source lines 4478–4478. Cross-chapter pointer whose target carries the canonical tag 03F3 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DDW: sources/stacks-chow/tags.txt, source lines 16781–16781. Cross-chapter pointer whose target carries the canonical tag 0DDW in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DFM: sources/stacks-chow/tags.txt, source lines 16842–16842. Cross-chapter pointer whose target carries the canonical tag 0DFM in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DFH: sources/stacks-chow/tags.txt, source lines 16837–16837. Cross-chapter pointer whose target carries the canonical tag 0DFH in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 09XN: sources/stacks-chow/tags.txt, source lines 12536–12536. Cross-chapter pointer whose target carries the canonical tag 09XN in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03CG: sources/stacks-chow/tags.txt, source lines 4383–4383. Cross-chapter pointer whose target carries the canonical tag 03CG in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03A0: sources/stacks-chow/tags.txt, source lines 4291–4291. Cross-chapter pointer whose target carries the canonical tag 03A0 in

the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 04K2: sources/stacks-chow/tags.txt, source lines 5958–5958. Cross-chapter pointer whose target carries the canonical tag 04K2 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 089J: sources/stacks-chow/tags.txt, source lines 10499–10499. Cross-chapter pointer whose target carries the canonical tag 089J in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DES: sources/stacks-chow/tags.txt, source lines 16812–16812. Cross-chapter pointer whose target carries the canonical tag 0DES in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $\mathcal{F}$ be a sheaf on $X_{étale}$. Then $\pi_X^{-1}\mathcal{F}$ is given by the rule*

$$(\pi_X^{-1}\mathcal{F})(Y) = \Gamma(Y_{étale}, f_{small}^{-1}\mathcal{F})$$

for $f : Y \to X$ in $(Spaces/X)_{étale}$. Moreover, $\pi_Y^{-1}\mathcal{F}$ satisfies the sheaf condition with respect to smooth, syntomic, fppf, fpqc, and ph coverings.

Proof. Since pullback is transitive and $f_{small} = \pi_X \circ i_f$ (see above) we see that $i_f^{-1}\pi_X^{-1}\mathcal{F} = f_{small}^{-1}\mathcal{F}$. This shows that π_X^{-1} has the description given in the lemma.

To prove that $\pi_X^{-1}\mathcal{F}$ is a sheaf for the ph topology it suffices by Topologies on Spaces, Lemma 0DFM to show that for a surjective proper morphism $V \to U$ of algebraic spaces over X we have $(\pi_X^{-1}\mathcal{F})(U)$ is the equalizer of the two maps $(\pi_X^{-1}\mathcal{F})(V) \to (\pi_X^{-1}\mathcal{F})(V \times_U V)$. This we have seen in Lemma 0.3.

The case of smooth, syntomic, fppf coverings follows from the case of ph coverings by Topologies on Spaces, Lemma 0DFH.

Let $\mathcal{U} = \{U_i \to U\}_{i\in I}$ be an fpqc covering of algebraic spaces over X. Let $s_i \in (\pi_X^{-1}\mathcal{F})(U_i)$ be sections which agree over $U_i \times_U U_j$. We have to prove there exists a unique $s \in (\pi_X^{-1}\mathcal{F})(U)$ restricting to s_i over U_i. Case I: U and U_i are schemes. This case follows from Étale Cohomology, Lemma 09XN. Case II: U is a scheme. Here we choose surjective étale morphisms $T_i \to U_i$ where T_i is a scheme. Then $\mathcal{T} = \{T_i \to U\}$ is an fpqc covering by schemes and by case I the result holds for $\mathcal{T}$. We omit the verification that this implies the result for $\mathcal{U}$. Case III: general case. Let $W \to U$ be a surjective étale morphism, where W is a scheme. Then $\mathcal{W} = \{U_i \times_U W \to W\}$ is an fpqc covering (by algebraic spaces) of the scheme W. By case II the result hold for $\mathcal{W}$. We omit the verification that this implies the result for $\mathcal{U}$. □

Lemma 0.2. *Let S be a scheme. Let $\tau \in \{étale, fppf, ph\}$ (add more here). The inclusion functor*

$$(Sch/S)_\tau \longrightarrow (Spaces/S)_\tau$$

is a special cocontinuous functor (Sites, Definition 03CG) and hence identifies topoi.

Proof. The conditions of Sites, Lemma 03A0 are immediately verified as our functor is fully faithful and as every algebraic space has an étale covering by schemes. □

Lemma 0.3. *Let S be a scheme. Let $f : Y \to X$ be a surjective proper morphism of algebraic spaces over S. Let $\mathcal{F}$ be a sheaf on $X_{étale}$. Then $\mathcal{F} \to f_*f^{-1}\mathcal{F}$ is injective with image the equalizer of the two maps $f_*f^{-1}\mathcal{F} \to g_*g^{-1}\mathcal{F}$ where g is the structure morphism $g : Y \times_X Y \to X$.*

Proof. For any surjective morphism $f : Y \to X$ of algebraic spaces over S, the map $\mathcal{F} \to f_*f^{-1}\mathcal{F}$ is injective. Namely, if $\overline{x}$ is a geometric point of X, then we choose a geometric point $\overline{y}$ of Y lying over $\overline{x}$ and we consider

$$\mathcal{F}_{\overline{x}} \to (f_*f^{-1}\mathcal{F})_{\overline{x}} \to (f^{-1}\mathcal{F})_{\overline{y}} = \mathcal{F}_{\overline{x}}$$

See Properties of Spaces, Lemma 04K2 for the last equality.

The second statement is local on X in the étale topology, hence we may and do assume Y is an affine scheme.

Choose a surjective proper morphism $Z \to Y$ where Z is a scheme, see Cohomology of Spaces, Lemma 089J. The result for $Z \to X$ implies the result for $Y \to X$. Since $Z \to X$ is a surjective proper morphism of schemes and hence a ph covering (Topologies, Lemma 0DES) the result for $Z \to X$ follows from Étale Cohomology, Lemma 0DDW (in fact it is in some sense equivalent to this lemma). □

Lemma 0.4. *Let S be a scheme. Let X be an algebraic space over S.*

1. *For $\mathcal{F} \in Sh(X_{étale})$ we have $\epsilon_{X,*}a_X^{-1}\mathcal{F} = \pi_X^{-1}\mathcal{F}$ and $a_{X,*}a_X^{-1}\mathcal{F} = \mathcal{F}$.*
2. *For $\mathcal{F} \in Ab(X_{étale})$ torsion we have $R^i\epsilon_{X,*}(a_X^{-1}\mathcal{F}) = 0$ for $i > 0$.*

Proof. We have $a_X^{-1}\mathcal{F} = \epsilon_X^{-1}\pi_X^{-1}\mathcal{F}$. By Lemma 0.1 the étale sheaf $\pi_X^{-1}\mathcal{F}$ is a sheaf for the ph topology and therefore is equal to $a_X^{-1}\mathcal{F}$ (as pulling back by ϵ_X is given by ph sheafification). Recall moreover that $\epsilon_{X,*}$ is the identity on underlying presheaves. Now part (1) is immediate from the explicit description of π_X^{-1} in Lemma 0.1.

We will prove part (2) by reducing it to the case of schemes – see part (1) of Étale Cohomology, Lemma 0DE4. This will "clearly work" as every algebraic space is étale locally a scheme. The details are given below but we urge the reader to skip the proof.

For an abelian sheaf $\mathcal{H}$ on $(Spaces/X)_{ph}$ the higher direct image $R^p\epsilon_{X,*}\mathcal{H}$ is the sheaf associated to the presheaf $U \mapsto H^p_{ph}(U, \mathcal{H})$ on $(Spaces/X)_{étale}$. See Cohomology on Sites, Lemma 072W. Since every object of $(Spaces/X)_{étale}$ has a covering by schemes, it suffices to prove that given U/X a scheme and $\xi \in H^p_{ph}(U, a_X^{-1}\mathcal{F})$ we can find an étale covering $\{U_i \to U\}$ such that ξ restricts to zero on U_i. We have

$$\begin{aligned} H^p_{ph}(U, a_X^{-1}\mathcal{F}) &= H^p((Spaces/U)_{ph}, (a_X^{-1}\mathcal{F})|_{Spaces/U}) \\ &= H^p((Sch/U)_{ph}, (a_X^{-1}\mathcal{F})|_{Sch/U}) \end{aligned}$$

where the second identification is Lemma 0.2 and the first is a general fact about restriction (Cohomology on Sites, Lemma 03F3). Looking at the first paragraph and the corresponding result in the case of schemes (Étale Cohomology, Lemma 0DDW) we conclude that the sheaf $(a_X^{-1}\mathcal{F})|_{Sch/U}$ matches the pullback by the "schemes version of a_U". Therefore we can find an étale covering $\{U_i \to U\}$ such that

our class dies in $H^p((Sch/U_i)_{ph}, (a_X^{-1}\mathcal{F})|_{Sch/U_i})$ for each i, see Étale Cohomology, Lemma 0DE4 (the precise statement one should use here is that V_n holds for all n which is the statement of part (2) for the case of schemes). Transporting back (using the same formulas as above but now for U_i) we conclude ξ restricts to zero over U_i as desired. □